

Let's Sleep On It:
Measuring the Impact of Sleep in Consolidating Prior Learning
with the Brain App –
a Novel, Personalized, Non-invasive App to Quantify Learning, Available Globally

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Abstract

Despite the recent remarkable advances in Artificial Intelligence (AI), humans still learn orders of magnitude faster (Tsividis et al., 2021). Our brains have evolved to figure out hidden structures of problems and use this knowledge to generalize across similar problems, enabling complex rapid inferencing even when limited to sparse data (Behrens et al., 2018). Recent ground-breaking discoveries suggest that spatial relational inferences in the brain are based on internal representations of the world as two-dimensional cognitive maps (Moser et al., 2015). These internal models are supported by the neural mechanisms in the *entorhinal hippocampal system* including *place cells* and *grid cells* (Hafting et al., 2005). Moreover, these cells spontaneously recapitulate old and explore new spatial trajectories during periods of slow wave sleep – a phenomenon known as *replay* (Ólafsdóttir et al., 2015). Replay is known to play a critical role in consolidating learned structure of spatial tasks and enabling generalization, a benefit that may operate below the level of conscious awareness (Liu et al., 2021). More recent studies in humans suggest that the mechanisms for constructing the spatial cognitive map may be examples of a more general coding mechanism capable of building cognitive maps covering any nonspatial or abstract domain (Behrens et al., 2018; Eichenbaum 2017; Epstein et al., 2022). If so, could rest or sleep – associated with replay – also preferentially facilitate the learning of structure in non-spatial abstract tasks thereby allowing superior performance for inferencing? How could one quantify the impact of sleep on inferencing in abstract tasks in general environments on a mass scale completely outside of lab settings without the burden of invasive or expensive apparatus?

In this work, I aim to answer both these questions. First, I built a Cloud-native, software application, The Brain App, which invites users to play an interactive “Card Memory Game” with scheduled learning and testing phases separated by a time delay. In the testing phase, the app scores the users’ mental model of the structure of relationships experienced in the learning phase. Second, I employed the The Brain App to measure learning among a cohort of 68 human subjects divided into two groups: the Diurnal (Control) Group vs. the Nocturnal (Manipulated) Group. The differences in the aggregate learning metrics between two groups provide the first-ever experimental evidence out of the lab environment that indeed, sleep facilitates the learning of structure in non-spatial abstract tasks with improved inferencing, perhaps giving some adage to the phrase “*Let’s sleep on it.*”

My work provides a general end-to-end framework for personalized non-invasive cognitive metrics to quantify learning in any human subject. The Brain App can be tuned to monitor cognitive health in a neurodegenerative patient or to track learning of arithmetic among elementary students. Uncovering the *Code of Human Thought* is one of the greatest scientific quests ever undertaken, and the current and future work with The Brain App will play a role in this quest to understand the basis of cognition and to help engineer cognition in artificial intelligence systems.

Introduction

Our brains have evolved to figure out hidden structures of problems and use this knowledge to generalize across similar problems, thereby enabling complex rapid inferencing even when limited to sparse data (Behrens et al., 2018). For example, different spatial environments share the common regularities of Euclidean space, where we “know” moving n steps in each direction in the sequence north→east→south→west, irrespective of the magnitude n , will always bring us back to where we started. Similarly, in a non-spatial scenario, upon observing a sequence of tennis matches where Alice beats Bob, and Bob beats Carol, and Carol beats Dave, we can infer that Alice is likely to beat Dave. Thus, through the process of interrelating this information upon a linear ordering of the relationships between the tennis players, as if on a one-dimensional coordinate system, we can make inferences even though we may not have observed Alice playing with Dave.

Recent ground-breaking discoveries in cognitive neuroscience are beginning to provide insights into *how*, *where*, and *when* these fundamental properties of human cognition including memory, knowledge representation, learning and inferencing takes shape in the brain. These findings suggest that spatial relational inferences in the brain are based on internal representations of the world as two-dimensional *cognitive maps* (Moser et al., 2015). These internal models are supported by the neural mechanisms in the Entorhinal Hippocampal System including place cells and grid cells (Behrens et al., 2018; Hafting et al., 2005; O'Keefe et al., 1978). These cells spontaneously recapitulate old and explore new spatial trajectories during periods of slow-wave sleep – a phenomenon known as replay. In replay, prior paths traversed by an animal can be retraced 20 times faster than the real traversal of the path. Replay is known to play a critical role in consolidating learned structure of spatial tasks and enabling generalization, a benefit that may operate below the level of conscious awareness (Liu et al., 2021).

New research has shown that, presumably driven by evolution, neuronal parallels exist between both spatial and abstract (non-spatial) cognition, allowing abstract concepts to be organized upon a map-like-representations used in spatial navigation for tasks beyond spatial navigation – a cognitive map – for more general cognitive domains such as abstract memory, learning, and decision making (Constantinescu et al., 2016).

If replay is also at play in non-spatial cognition, it might provide humans with abstract reasoning and generalization capabilities far exceeding those of other mammals. If so, could rest or sleep – often associated with replay – also preferentially facilitate the learning of structure in non-spatial abstract tasks thereby allowing superior performance for inferencing? How could one quantify the impact of sleep on inferencing in abstract tasks in general environments on a mass scale completely outside of lab settings without the burden of invasive and/or expensive apparatus?

This study addresses both these research questions in inviting human subjects to play an interactive game in The Brain App which then scores the user's innate ability to learn and build a mental model of the hidden structure of the game.

Review of Literature

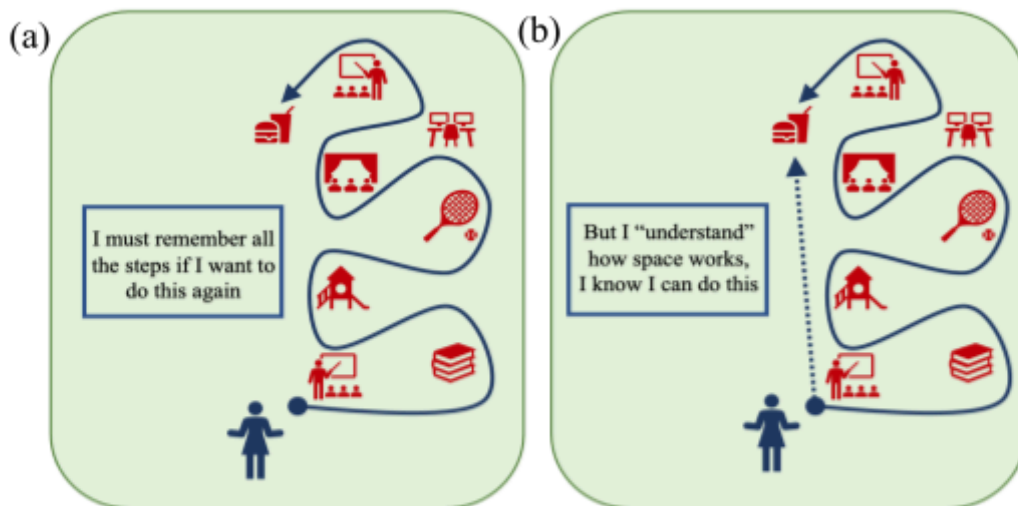


Figure 1. (a) A kindergartener on her first day at school taking a tortuous route from her classroom to the cafeteria. To get to the cafeteria from her classroom the next day, she has to remember the sequence of landmarks in her route. (b) But if she “understands” that the campus landmarks are in a two-dimensional space, she can infer that there could be a “shortcut” from her classroom to the cafeteria. Recent studies suggest that the brain can spontaneously recapitulate prior routes and explore alternative paths in periods of rest and sleep giving rise to the possibility that such understanding or intuition can emerge involuntarily from below the level of conscious awareness (Adapted from BRI UCLA Joint Seminar In Neuroscience by Timothy Behrens, 2022).

Spatial Cognition

Imagine a kindergartener finding a torturous route from her classroom to the cafeteria (Figure 1a). If the student wants to go to the cafeteria again, she has to take the same set of actions and the same torturous route, but if she understands that space is a special two-dimensional topology, and understands the structure of space, then she can make a much shorter route next time (Figure 1b). These kinds of spatial inferences spontaneously engage in such learning everyday as we drive or walk the same route daily and learn the relative locations of various landmarks on the way (Tolman et al., 1948).

Recent discoveries have shed light on the neurons in our brain that underlie these inferences in the spatial domain (Figure 2, overleaf). Through analyzing neuronal activations during navigational tasks in mammals (Chang et al., 2017; Epstein et al., 2017; Fenton et al., 2020; Moser et al., 2015), scientists have articulated the brain's spatial navigation system. Localized neurons in the *entorhinal hippocampal system* (EHS) have been found to encode navigational variables. Together the spatially tuned cells fire in a way that internally mirrors the spatial geometry of the surrounding environment. This internal representation of our environment has been coined as a *cognitive map* (Moser et al., 2014).

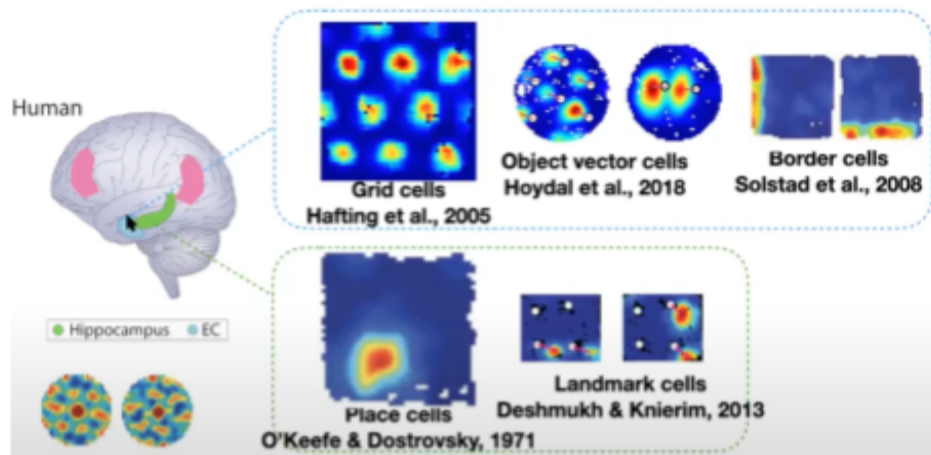


Figure 2. Key spatially tuned cells in the hippocampal entorhinal system (BRI UCLA Joint Seminar In Neuroscience by Timothy Behrens, 2022).

Grid Cells. At the cellular level, the entorhinal cortex is made up of neurons called *grid cells* that are crucial to composing the cognitive map. Grid cells act as rulers, signaling the distance between locations. Grid cells are internally arranged in grids of repeating equilateral triangles tiling the spatial environment. This triangular grid arrangement resembles a coordinate system, and allows for the distance between spatial landmarks to be calculated (Figure 2) (Bellmund et al., 2018).

Place Cells. Alongside the grid cells, *place cells* located in the hippocampus support the cognitive map by signaling the animal's position. *Place field* activation maps show place cell firings in path-like trajectories, akin to the actual path being traversed by the individual, suggesting that place cells represent an animal's current and past locations by mirroring spatial trajectories upon our internal model—our cognitive map—of the world (Figure 2) (Bellmund et al., 2018). Every place cell has a different preferred firing location within the animal's local environment, meaning that each location can be mapped to a specific place cell. This feature allows place fields to generate map-like representations of our surroundings. Thus, while grid cells and place cells both contribute spatial information necessary in composing the cognitive map, these spatially tuned cells govern different navigational abilities. Place cells represent specific locations in space, and our paths within such spaces while grid cells measure the distance between locations.

Replay

Another key feature of spatial cognition is the ability to *replay* past trajectories in space. Recording studies in rodents have suggested that place cells sequentially activate during periods of sleep, mirroring past spatial trajectories traversed on a maze (Epsztein et al., 2022) (Figure 3a). This phenomenon termed as “replay” demonstrates the ability of spatially tuned cell types to represent

locations outside of the animal's current location, for example previous paths in a maze. Place cell sequential activation exhibited during replay are temporally compressed (Figure 3b). This compression allows for faster simulations of previous spatial trajectories along the cognitive map.

Replay is essential to spatial cognition as it allows for the simulations of traversing possible paths, coupled with the potential outcomes associated with that decision (for example taking a turn in a maze and being met with a dead end). This ability to test out trajectories supports more informed decision making in future tasks (Bellmund et al., 2018).

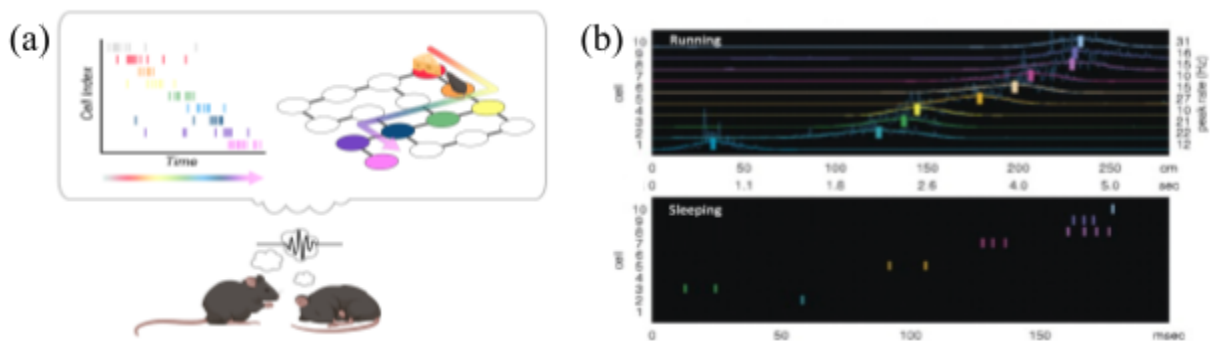


Figure 3. (a) Spatial replay of past trajectories during resting periods (Skaggs et al., 1996). (b) Hippocampal cellular activation during periods of spatial traversal across a five second span, and the replay of such spatial trajectories during sleeping periods across a 200 millisecond span. (Lee and Wilson et al., 2001).

Structural Inferences

Space isn't the only problem that the EHS engages in. Problems with transitive relationships also involve using the structure of the world to make an inference (Whittington et al., 2020). For example if we know that A is greater than B and that B is greater than C, we can then make the inference that A is greater than C despite never having learned this information directly. Such inferences emerge when a hidden underlying structure is identified, in this case the "greater than relationship" structure which lies on a one dimensional topology. Equivalently, in observing a sequence of tennis matches where Alice beats Bob, and Bob beats Carol, we can interrelate these two facts and infer that Alice is likely to beat Carol. Thus, through the process of interrelating this knowledge upon an underlying structure, knowledge can be inferred beyond the individual component facts (Figure 4, overleaf) (Smith et al., 2005). How and when the brain finds the structure of such problems is a key motivation of this current study.

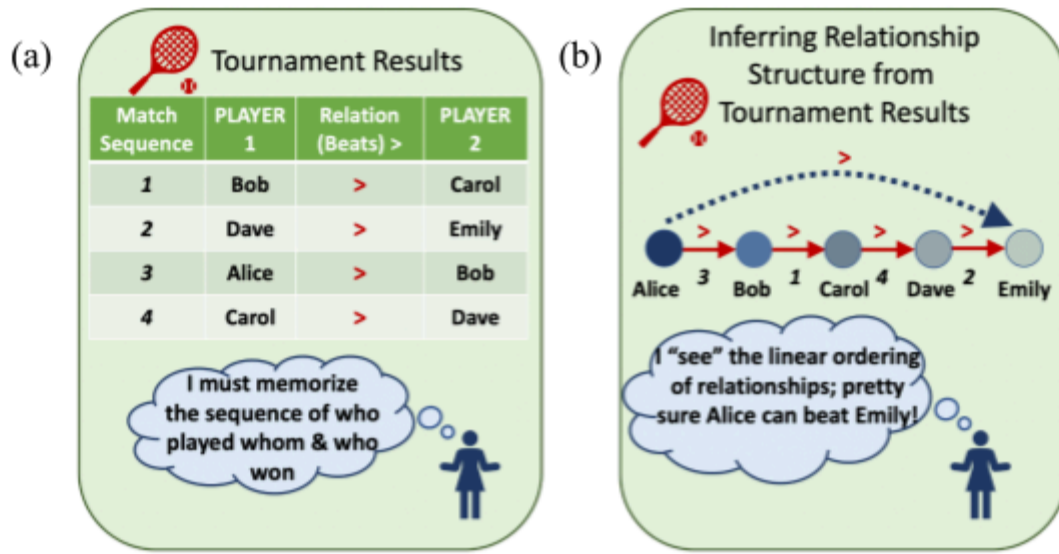


Figure 4. (a) A kindergartener trying to memorize the results of a sequence of singles matches in a tennis tournament. **(b)** But, if she “understands” the linear (and one-dimensional) ordering of the pairwise relationships as to who-beat-whom, she can infer that Alice is likely to beat Emily even though Alice has not played with Emily. If replay is also at play in non-spatial cognition, it might provide humans with abstract reasoning and generalization capabilities far exceeding those of other rodents. Adapted from BRI UCLA Joint Seminar In Neuroscience by Timothy Behrens (2022).

Representing Relationships

A set of relationships can be represented as a graph whereby each node on the graph is a sensory stimulus that is experienced, say, observing the shape of the moon. Then, an edge between two sensory stimuli represents a transition (or relation) between the two sensory states; a full moon will be followed by a waning gibbous moon. Such graphs can take up different structural forms including circular (e.g., the phases-of-the-moon graph, the cycle-of-seasons graph and the day–night cycle), hierarchical (e.g., a family tree, a management chain in a workplace, a set of files stored in folders with sub-folders), community structure (e.g., the social network graph), or two-dimensional spatial environment by respecting transition/relational structure that respects the translational and rotational invariances of two-dimensional space. Humans can extract abstract information and use it to facilitate new inferences. Furthermore, the EHS is argued to represent this knowledge efficiently (Mark et al., 2020).

Inferences about structures underlying tasks potentially offer advantages in learning when encountering a novel environment. Further, relationships that have never been observed can be inferred because the structure of the problem is understood (Tenebaum et al., 2011). Two-dimensional maps are formed in the EHS that represent our physical environment, and such maps allow us to infer the spatial relationships between landmarks in an environment.

Modern cognitive neuroscience has shown that the human brain employs the architecture of map-like representations used in the EHS for spatial navigation to tasks beyond spatial navigation: a cognitive map for broader cognitive domains. The brain organizes concepts into a mental map, allowing conceptual relationships to be navigated in a manner similar to spatial relationships; the relationship between concepts are measured similarly to how the distance between environmental landmarks are measured in a cognitive map (Aronov et al., 2017)

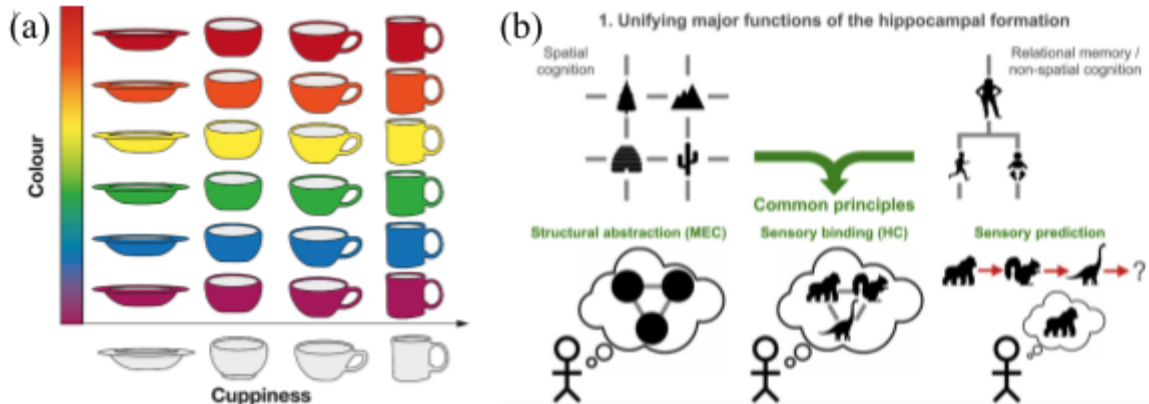


Figure 5. (a) Different objects may be factorized according to their “cuppiness” and “colour” (Behrens et al., 2008). (b) Unifying major functions between spatial and nonspatial cognition (Whittington et al., 2020).

The brain represents abstract concepts and their relationships such as knowledge of algebra, hierarchies of friendships, or the representation of different types of ceramic tableware upon an internal map—our cognitive map. Consider the following example; when searching through your cabinet, you might subconsciously describe the ceramic tableware along two-dimensions, color and cuppiness (Figure 5a). Depending on the two features, the ceramic tableware can be located anywhere within this two-dimensional space, similar to the way that physical space is represented on a map. This representation of abstract concepts is encoded onto our cognitive maps, supported by the same grid cells and place cells used in spatial navigation (Figure 5b) (Bellmund, 2018; Rueckemann et al., 2021).

Most research on measuring abstract cognition is limited to invasive approaches (e.g. brain lesion studies, electrical recordings of neuronal activations) or expensive apparatus (e.g. Magnetoencephalography studies), both of which are not scalable for adoption among the general public. However in spatial cognition, one approach that has recently gained prominence is path integration tasks (Stangl et al., 2018) .

Measuring Spatial Cognition. *Path integration* is a process by which an individual uses information generated from their own body movement, such as distance and orientation, to keep track of their location upon a two-dimensional space (Figure 6a, overleaf). Path integration in humans is a spatial

ability implicated within the properties of the EHS, thus making it a good measure of spatial cognition. (McNaughton et al., 1996). The discovery of these path integration tasks have been leveraged by the virtual reality app called Sea Hero Quest, whose mission was to measure the effectiveness of cognitive processes supporting navigation in two-dimensional space. After successfully gathering data from 4.3 million players, aggregated population-level Sea Hero Quest data is now available for scientists for research into cognition and spatial navigation (Figure 6b) (Spiers et al., 2021).

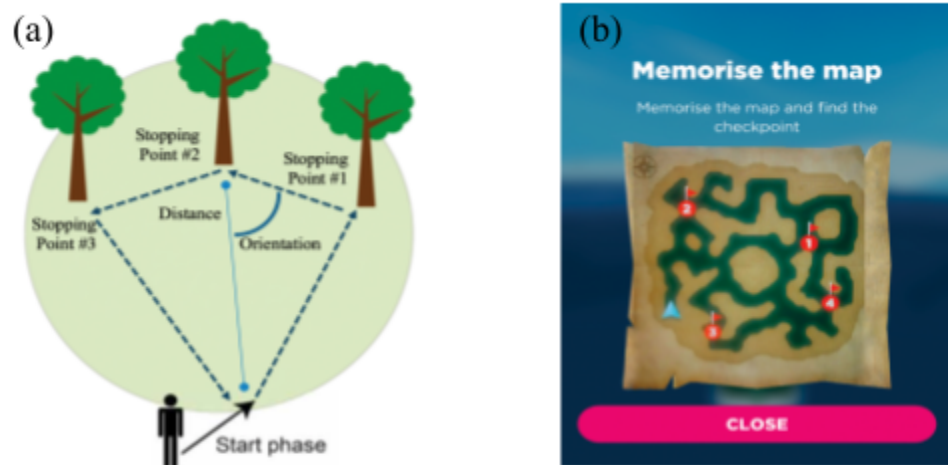


Figure 6. (a) In this example of path integration, there are three stopping points along the path. While walking along the path blindfolded, the individual must stop at each stopping point and estimate the distance and orientation relative to the path starting point (Bierbrauer et al., 2020). (b) Virtual Navigation on a mobile Sea Hero Quest app.

Measuring Abstract Cognition. There is an opportunity to recognize these new advances in cognitive neuroscience, and therefore probe the representation of abstract knowledge in the brain. In this current study, The Brain App is developed to measure abstract cognition, akin to path integration as a mechanism to measure spatial cognition.

Existing literature suggests that replay exists in spatial contexts, however replay in abstract cognition is still an area of active research. In this study, The Brain App is implemented to non-invasively investigate the relationships between abstract structural knowledge and short wave sleep – often associated with replay.

Project Overview

This study investigates the impact sleep has on the ability to consolidate prior learning in abstract/non-spatial tasks. The Brain App invites the user to play an interactive “Card Memory Game” where sequences of pairs of cards (e.g., Kiwi → Seashell), representing pairwise relationships, are presented in the learning phase. The sequence of pairs of cards are drawn repeatedly, with replacement, in

a randomized order from an underlying graph of relationships which is never presented to the user. Later, in the testing phase, The Brain App scores the user's innate ability to learn and build a mental model of the yet-unseen structure of the underlying graph through different sets of queries such as "Which (card) is Closer" or "Navigate to the target (card)", given a random starting card. By managing the duration and the nature of the delay between the learning and testing phases, The Brain App provides a principled, practical and quantitative basis to measure learning across different time points for any user using the app from the computing device of their choice.

Objectives

1. Design and develop a platform –The Brain App– to measure a user's innate ability to learn and build a mental model of the unseen structure of underlying graphs.
2. Investigate the impact of sleep – often associated with replay – in preferentially facilitating the learning of structure in non-spatial abstract tasks thereby allowing superior performance for inferencing

Hypotheses

H_{01} : There will be no difference in the percent accuracy during Distance Estimation and Navigation tasks between the Diurnal Group and the Nocturnal Group.

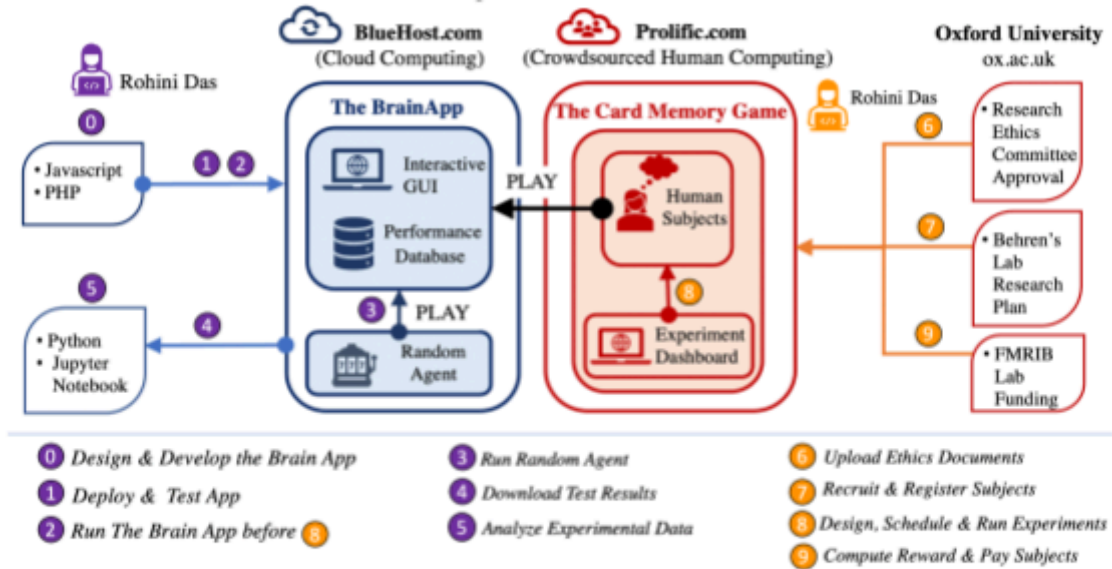
Alternative Hypothesis: The Nocturnal Group will perform with higher percent accuracy in the Distance Estimation and Navigational tasks when compared to the Diurnal Group.

My Role In Study

Initially, I became interested in cognitive neuroscience as I was inspired by the 2014 Nobel Prize work in Physiology and Medicine for its groundbreaking discovery of spatially tuned neurons in the EHS governing navigation. In the first year of my enrollment in the nationally recognized three-year science research program at Byram Hills High School (BHHS), I reached out to Dr. Tim Behrens from Oxford University after independently studying literature relating this Nobel Prize work to discoveries in human replay.

Throughout the next year (September 2021 through August 2022), I built my computer science skills from the ground up to design and implement The Brain App in JavaScript and HTML and then deployed it to run on Bluehost.com's Cloud Computing Servers in the US. With the help of my mentors, I received Oxford University Research Ethics Committee's approval to conduct the experiments with human subjects for my study (CUREC Approval Reference: R79126/RE001). With the approval in hand, I personally recruited over one hundred human subjects from Prolific.com, a crowdsourced human-computing website in the UK and registered each of the participants to play the *Card Memory Game*. Next, I placed participants into equal sized control and manipulated groups and messaged each

participant to confirm their scheduled sessions for learning and testing, while also responding to their questions before, during and after their sessions. I ran the experiment and computed the bonus reward of each participant, according to their performance, and compensated them each individually with the Functional MRI of the Brain (FMRIB) lab funding. I downloaded the results in the database on BlueHost and analyzed the experimental data on Python in Jupyter Notebooks (Figure 7).



Figures 7. Thematic representation of my roles in this study. The left side (colored in purple) highlights my six key activities(0-5) related to The Brain App running on BlueHost.com, a Cloud Computing site in the US. Similarly, the right side (colored in orange) highlights my four key activities related to conducting the experiments in Prolific.com, a Crowdsourced Human Computing site in the UK. My study was approved by Oxford University Research Ethics Committee. The design and schedule of the experiments followed guidelines from the Behren's Lab at Oxford University, and the FMRIB Lab, Oxford University provided funds to reward and pay human subjects participating in the study.

Methods

Behavioral Experiment

In this experiment, 68 human subjects were recruited. 33 subjects were assigned to the Diurnal Group and 35 subjects were assigned to the Nocturnal Group. All the subjects were UK citizens between the age of 18 and 35. After sharing the experimental procedure and task description, consent forms were given to each subject prior to engaging in the experiment. This study was approved by the Central University Research Ethics Committee, UK.

Experimental Procedure

This experiment consisted of two sessions, a learning session, and a testing session. Participants were segregated into two groups (Diurnal and Nocturnal). The Diurnal Group was the control group; their learning session was on Day 1 at 8:00 am and their testing session was later that day (Day 1) at 8:00 pm.

The Nocturnal Group was the manipulated group; their learning session was on Day 1 at 8:00 pm and their testing session was the next morning (Day 2) at 8:00 am, allowing for a period of nocturnal sleep between the learning and testing sessions that was not offered to the Diurnal Group (Figure 8).

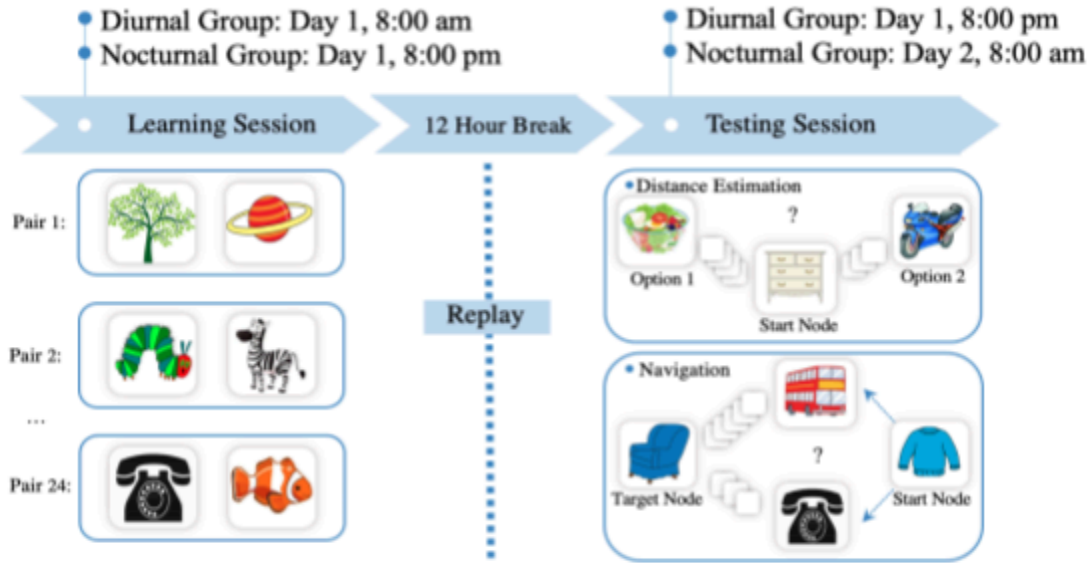


Figure 8. Unlike the diurnal group, the nocturnal group participants were provided with a period of sleep between their learning and testing sessions.

The sleep period in the nocturnal group is intended to be analogous to the resting periods in rodent studies where hippocampal replay has been observed in prior spatial tasks (Davidson et al., 2009; Foster et al., 2006). This experimental procedure was inspired by the measurement of replay in humans using Magnetoencephalography (Liu et al., 2019). In this study, we use The Brain App to non-invasively and inexpensively measure the relationship between non-spatial tasks and hippocampal replay.

Data Collections

Once the experiment was published on Prolific, an Oxford-based Crowdsourced Human Computing company (Figure 9, overleaf), participants were recruited and paid at a set rate of £9 an hour. To give the incentive to be attentive during the learning session, participants were informed that the better they performed during the testing session, the higher they would be compensated with a reward.

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Figure 9: Prolific page for the nocturnal group participants signing up for the learning session.

Task Design

Following the approaches of Kemp and Mark, this study formalizes the inferences over structural forms using generative models of graphs (Kemp et al., 2008; Mark et al., 2020). In The Brain App, the observations of the participants are Markovian (each state depends only on the state before) and follow a transition matrix that is defined by an underlying graph structure (for example, grid, hexagonal, or community) with given parameters such as the size of the graph, including the number of nodes and edges in the graph (Figure 10).

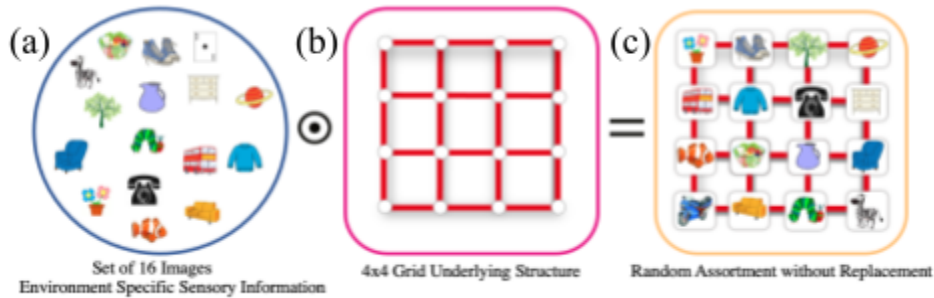


Figure 10. (a) Set of 16 images (nodes) that will be used in the graph. These images make up the environment specific sensory information. (b) 4x4 grid underlying structure. (c) Random assortment without replacement.

Furthermore, each node is associated with a sensory stimulus, represented in this study by an image. Thus, the total number of distinct sensory stimuli has to match the number of nodes in the graph.

For each participant, the assignment of sensory stimuli to the nodes is made at random without replacement, and is fixed throughout the duration of the learning phase and the testing phase.

In this current study, subjects learned a graph consisting of 16 nodes with an underlying grid structure.

Learning Session

In the learning session, “Learn Random Pairs” (LRP), participants learn the graph through observing pairs of neighboring stimuli on the graph. To make the task more appealing, this graph learning task was presented as the “*Card Memory Game*”. In introducing the game for the learning phase of the experiment, participants were asked to imagine there was a deck of cards with pictures of everyday items (Figure 11a). The deck is then thrown onto the floor, leaving the cards randomly assorted across the floor. On the arrangement of cards on the floor, a web is created that connects the cards together, as shown in red (Figure 11b). The objective of the Card Memory Game is to infer arrangement of cards on the floor without ever viewing the complete arrangement.



Figure 11. *Unscrambling Card Memory Game* description given to participants. (a) Deck of cards. (b) Random arrangement of cards on the floor. (c) Observing pairwise relationships (a teacup and a kiwi).

Participants are only allowed to observe two cards that are immediate neighbors of each other, connected by the web. For example, a pair of cards such as a kiwi and a teacup would be shared, indicating that they are neighboring each other on the web (Figure 11b). Participants were aware that the pairs presented were presented in an unordered fashion.

Discrete pairs of visual stimuli (cards) were provided as opposed to the viewing the entire graph (the floor arrangement of cards) so that in order to perform well during the testing session, participants needed to have reordered/unscrambled the individual pairs learned and infer a complete graph of the stimuli arrangement.

Testing Session

Twelve hours following the learning session, participants in both the Diurnal and Nocturnal Group were quizzed on their understanding of the card arrangements through two different path integration tests, *Distance Estimation* and *Navigation*.

Distance Estimation. During the first task, *Distance Estimation*, participants were to report which of the two cards presented to them appeared closer to the target node according to the arrangement they had previously learned in LRP (Figure 12a). For example, the participants could be asked if the cloud card (the Target Card) is closer to the juice box card (Card 1) or the scarf card (Card 2) (Figure 12a). In this sample question, the correct answer would be the scarf card because you can reach the cloud card from the scarf card in at least two edges while the cloud card is reached from the juice box card in at least 3 edges (Figure 12b). The Distance Estimation task consisted of 14 trials.

This task designed to measure non-spatial cognition is analogous to the spatial path integration task used to measure spatial cognition. Similarly to how participants estimated the spatial distance between their current position and their starting position along the trajectory, participants in this study estimated the distance between two cards; instead of navigating physical space, the participants are navigating abstract concepts.

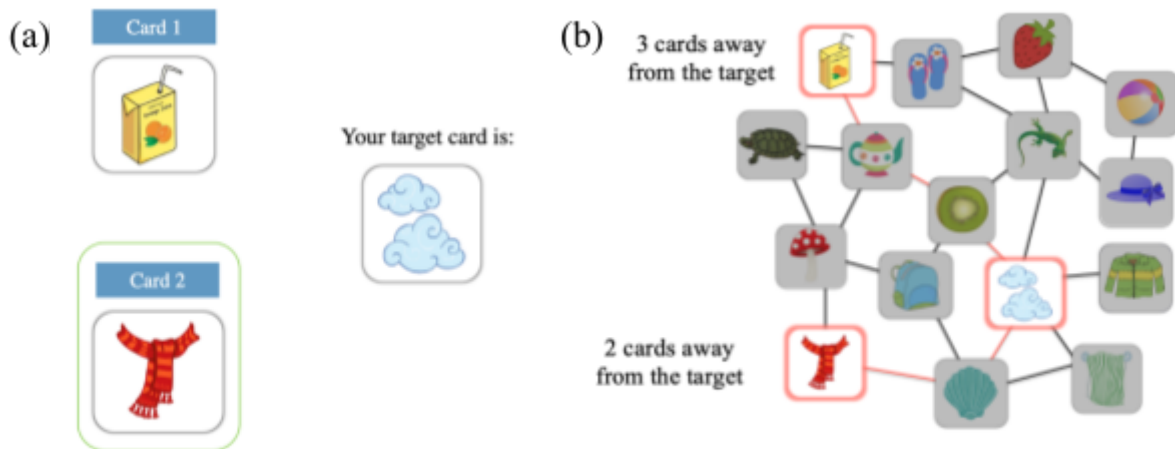


Figure 12. (a) Distance Estimation Task; “Which card (Card 1 or Card 2) is closer to the target card”. **(b)** Answer Explanation.

Navigation. During the second task, *Navigation*, participants traversed the card arrangement. Starting from the “current card” (for example the kiwi card), participants were instructed to select one of the two neighboring cards of the current card (the salamander card and backpack card) until they arrived at the target picture (the strawberry card) (Figure 13a, overleaf).

Every time the subject selected a card, their selected card then became the current card. The participants were not permitted to go to the next trial until they had successfully navigated to the target

card. The Navigation task consisted of 5 trials, and the minimum distance to the target node increased upon each successive trial. The minimum distance to the target was two cards in the first and second trials, three cards in the third and fourth trials, and four cards in the fifth and final trial.

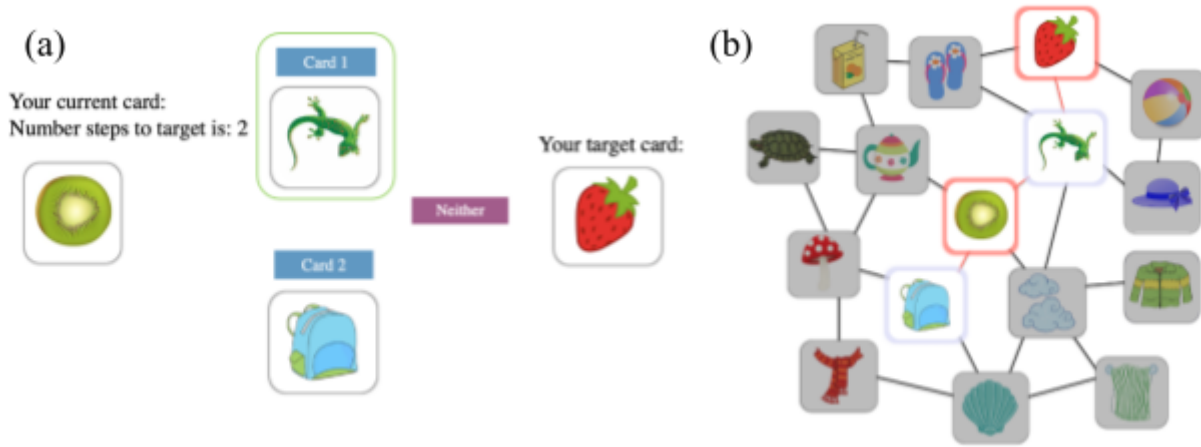


Figure 13. (a) Navigation Task: “Navigate from the current card to the target card by choosing between Card 1 and Card 2”. **(b)** Answer Explanation.

Random Agents

To facilitate the comparison of the performance of human subjects against a reference baseline, The Brain App was designed to include software-defined Random Agents which interact with the app through its graphical user interface (GUI), just as the human subjects. The Random Agents were built using the Python library Selenium. The choices made by the Random Agents for the two tasks in the testing phase can be tuned by the experimenter. In this study, the Random Agent always selected each option in each trial with equal probability. This simulation based approach quantifies the performance of agents with no learning ability.

Statistical Analysis

To analyze the *Distance Estimation* and *Navigation* task data, the data is visualized using RStudio. A significance value of $p < 0.05$ was used. To determine the significance of the difference in percent accuracy between Diurnal and Nocturnal Group in both tasks, a Welch Two Sample T-test was conducted as well as a right tailed test, where the alternative hypothesis tests whether the percent accuracy had increased from the Diurnal Group to the Nocturnal Group.

Results

In this study, the relationship between the consolidation of structural abstract knowledge and sleep was investigated. To accomplish this, a novel behavioral task was designed where participants acquired structural knowledge (card arrangements) and twelve hours later, participants were tested on their ability

to infer hidden structure from the unordered sequence of cards presented to them in the learning session through two metrics, Distance Estimation and Navigation.

Distance Estimation Task Performance

Results from a right tailed t-test showed a significant difference between the Diurnal and Nocturnal Groups' mean value of percent correct in the Distance Estimation task ($p = 0.032$) (Figures 14a). The Diurnal Group performed with an average percent correct value of 51.7% while the Nocturnal Group performed with an average percent correct value of 58.7%. These results suggest that participants who had a resting period between the learning session and the testing session were able to perform more successfully in the distance estimation task. Notably, this effect was driven by the manipulated variable between the two groups: sleep – often associated with hippocampal replay.

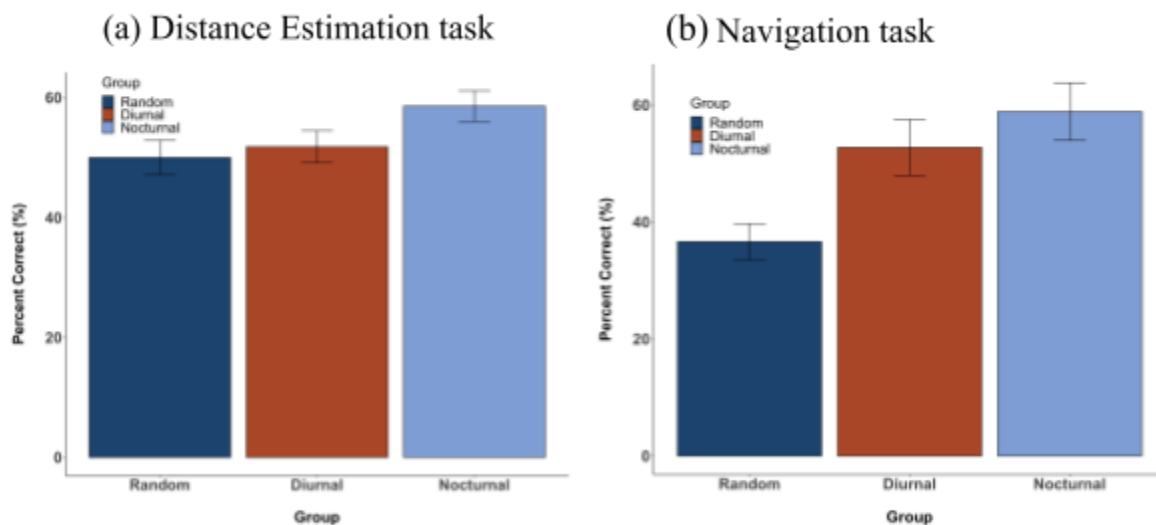


Figure 14. (a) Distance Estimation task performance after 14 trials for the Random Agent (theoretical assuming binomial distribution), Diurnal Group and Nocturnal Group as measured by mean percent correct. (b) Navigation task performance after 5 trials for the Random Agent (simulation), Diurnal Group and Nocturnal Group as measured by mean percent correct.

Navigation Task Performance

By the end of the 5 Navigation trials, the Random Agent performed with an average percent correct value of 35.6%, the Diurnal Group performed with an average percent correct value of 52.7% while the Nocturnal Group performed with an average percent correct value of 58.8%. The Nocturnal Group performed with a significantly greater average percent correct rate compared to the Diurnal Group ($p=0.009$) (Figure 14b).

The number of steps taken to reach the target was additionally used as a metric to judge performance. By the end of the 5 Navigational trials, the Random Agent took an average of 23 steps to reach the target card, the Diurnal Group took an average of 19 steps to reach the target card, and the Nocturnal Group took an average of 12 steps to reach the target card. Participants in the Nocturnal Group performed with a significantly lower average number of steps per trial in the Navigation task ($p=0.01$) (Figure 15).

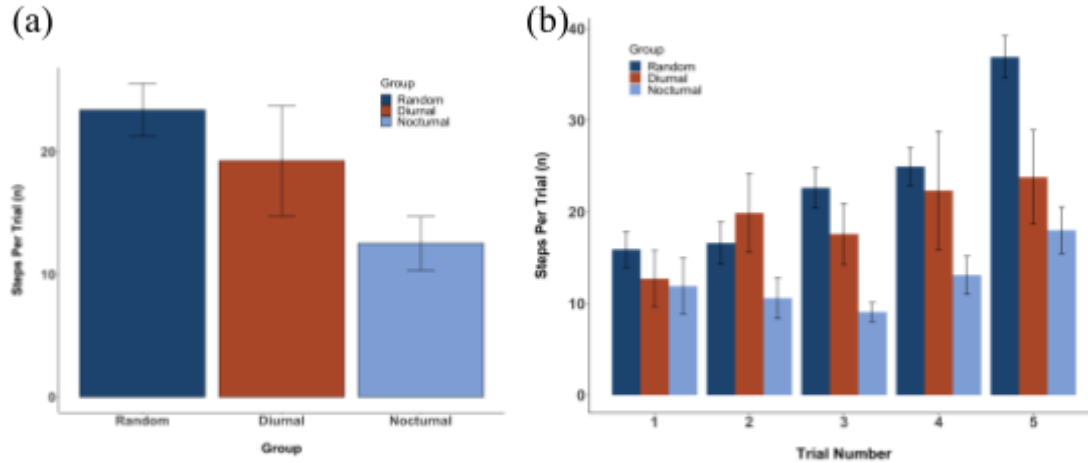


Figure 15. (a) Navigation task average steps per trial by the end of the 5 trials for the Random Agent Group, Diurnal Group, and Nocturnal Group. **(b)** Navigation task average steps per trial for the Random Agent Group, Diurnal Group, and Nocturnal Group.

Discussion

The ability to effortlessly solve problems by learning hidden structures embedded within the problems is a fundamental property of learning and one that allows humans to make rich inferences in novel situations. Here, we explore the role of sleep in the consolidation of prior learning and assisting in making transitive inferences.

Across the different quantitative metrics tested in The Brain App, the Nocturnal Group performed with significantly higher percent accuracy compared to the Diurnal Group. Notably, this difference is likely driven by the period of sleep which was the manipulated variable between the two groups, demonstrating that periods of sleep appear to be necessary in triggering the consolidation of prior learning through assisting in identifying the hidden structure of non-spatial tasks.

This study defines new cognitive metrics to quantify learning in *any* domain through measuring internal representations non-invasively. Building upon proposed neuronal parallels between spatial and abstract cognition (Behrens et al., 2018), this study suggests that replay, a phenomenon exhibited during periods of sleep, also helps learning in abstract domains and can serve as a measure of learning in any specified domain irrespective of the sensory modality.

There were challenges involved with making sure that the recruited participants abided by the regulations that had been set prior to conducting the research (e.g., spending the correct amount of time on the learning and testing phase, not engaging in the tasks before or after the scheduled times). Many participants were confused with the GUI or were inattentive with the tasks. I had to go back and refine the GUI such that the user experience was more friendly and game-like. Extending this research to a larger group of participants spanning different regions and different characteristics (e.g. age groups) can also provide more support and clarity.

Future studies should exercise the flexibility of The Brain App and by addressing different research goals such as transferring knowledge across problems with different underlying structures perhaps in the same or different sensory modalities. Future research should also look at expanding this research by testing participants after a larger delay between the learning and testing session (24 hours or 48 hours as opposed to 12 hours).

Applications

The Brain App's Implications in Future Research. The Brain App allows the researcher to personalize the cognitive health care for the specific learning abilities or disabilities of the patient and monitor their change in performance over time. In future research, the sensory modality could be changed, to include visual, auditory, and haptic stimuli. In this experiment, I arranged the stimuli along a grid underlying structure, however, stimuli can be arranged upon hierarchical clusters, a community underlying structure, a hexagonal underlying structure, or a graph extending beyond two-dimensions. The size of the graph can also be changed to test for a greater and smaller number of nodes/edges. In this experiment, I used “Learn Random Pairs” as my learning method, and “Distance Estimation” and “Navigation” as my testing method. However, in creating The Brain App, three other tasks were implemented: “Learn Random Walks”, “Extending Picture Sequences”, and “Can it be middle”. These supplementary tests were not applicable to this study as I was testing for replay, however in testing for other cognitive abilities, these tests would be just as useful. It is my hope that these features of The Brain App help it to be tunable to the desired goals of the researcher.

Student Learning implications. One of the key educational conundrums of our age is Bloom's 2 Sigma Problem which says that students receiving personalized tutoring perform two standard deviations better than students learning in classroom education (Nebra, 2016). In leveraging recent findings in cognitive neuroscience, such as human replay and the cognitive map (Moser et al., 2015), The Brain App provides personalized quantitative metrics both during the learning phase as well as suggests directed remedial guidance during testing. This type of testing is aligned with the representation that the brain develops using the spatially tuned cell types in the EHS.

Neurodegenerative Prognostic Metrics. Another field where The Brain App could have a revolutionary impact is cognitive health care, particularly in old age. Currently, the cognitive health care industry requires invasive multimillion-dollar tools, only available in hospitals. With The Brain App, patients will be able to monitor changes in cognitive abilities, which can then serve as non-invasive prognostic measures for neurodegeneration, specifically the diseases that impact the EHS.

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Conclusion

Past literature in Cognitive and Computational Neuroscience details neuronal parallels between both spatial and abstract cognition, pointing to a general neural coding system capable of organizing knowledge of all kinds (Epstein et al., 2017). The precise nature of these general coding mechanisms remains elusive, although there are recent interdependent hypotheses that call for the need of a general, modular, and scalable platform that helps in exploring these hypotheses in a structured robust fashion. In addressing this challenge, I built and implemented The Brain App, with qualitative learning metrics by leveraging established software engineering tools and libraries allowing the app to be tuned in research environments and labs to address specific research goals, as well as implementing novel methods informed by recent findings in spatial cognition. In this study, measure the impact of sleep in consolidating prior learning. It was found that during distance estimation and navigational tasks, participants who were provided with a period of sleep –often associated with hippocampal replay– were able to successfully integrate and re-order previously learned structural information. These results indicate that indeed, sleep facilitates the learning of structure in non-spatial abstract tasks with improved inferencing in the Card Memory Game.

This work, encompassing designing, deploying, and conducting studies on The Brain App, provides the basis towards personalized non-invasive cognitive metrics to quantify learning in any human subject. The Brain App provides the basis towards revolutionizing the educational and cognitive health industry by shaping every child's ability to learn, while in old age, providing novel prognostic measures for neurodegeneration through the monitorization of undermining cognitive abilities.

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